

William Jackson

Senior Software Engineer – Vue C# (.NET) Azure/AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer operating within **SaaS, fintech, and data-intensive platforms**, delivering resilient full-stack systems using **Vue.js (2–3), TypeScript (3–5), and C#/.NET / ASP.NET Core (2–8)** across cloud-native environments. Expertise spans **microservices, event-driven architecture, and secure API ecosystems**, consistently improving system reliability by over **40% through observability, idempotency, and backpressure strategies**. Proven in modernizing legacy platforms into scalable distributed systems, implementing **OpenTelemetry-driven monitoring**, and optimizing performance under high concurrency while ensuring secure, maintainable, and production-grade delivery.

SKILLS

- **Frontend** — **Vue.js (2–3), TypeScript (3.8–5.x), JavaScript (ES6+)**, HTML5, CSS3, Tailwind CSS, Bootstrap, component design systems, accessibility, state management, SPA routing, performance profiling
- **Backend & APIs** — **C# (7–12), .NET Framework (4.6–4.8), ASP.NET Core (2.1–8)**, REST APIs, microservices, background jobs, WebSockets, API versioning, rate limiting, idempotency, secure coding (OWASP-aligned)
- **Data, Messaging & Search** — **PostgreSQL (11–16)**, MySQL, SQL Server, **Redis (5–7)**, MongoDB, DynamoDB, **Kafka (2.x–3.x)**, RabbitMQ, AWS SQS/SNS, **Elasticsearch (6–8)**, OpenSearch, Redshift, BigQuery
- **Cloud & DevOps** — **AWS, Docker (20.x–26.x), Kubernetes (1.20–1.30)**, Helm, Argo CD, GitOps, **Terraform (1.0–1.7)**, CloudFormation, GitHub Actions, GitLab CI, Jenkins, Azure DevOps, CircleCI, Azure, GCP
- **Observability, Quality & Security** — **OpenTelemetry, Datadog, Prometheus, Grafana**, ELK/EFK, CloudWatch, Jest, Mocha, Chai, React Testing Library, Cypress, Playwright, Pact-style contract tests, **OAuth2, OpenID Connect, JWT, Okta, Auth0, AWS Cognito, RBAC/ABAC**, audit logging

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Delivered scalable **Vue.js (2–3) frontend architecture** with reusable component systems that reduced UI rework by over **35% across multiple product teams** while improving maintainability and consistency.
- Engineered **BFF and microservice layers using C#/.NET (5–9) and ASP.NET Core**, stabilizing frontend-backend contracts and eliminating integration fragility across evolving services.
- Designed robust **REST API ecosystems** with consistent error handling, validation, and pagination, significantly improving frontend predictability and reducing production defects.
- Implemented **event-driven pipelines using Kafka**, resolving duplicate processing issues through idempotent consumers and achieving over **99.9% processing accuracy**.
- Built resilient asynchronous workflows with **AWS SQS/SNS and DLQ strategies**, enabling safe retries and large-scale data backfills without impacting live traffic.
- Optimized **PostgreSQL query performance** using indexing and execution plan tuning, improving API latency by more than **45% on high-load endpoints**.
- Introduced **Redis-based caching layers** and throttling mechanisms, reducing database load during peak traffic conditions by over **30%**, with consistent caching patterns applied across **AWS ElastiCache and Azure Cache for Redis** deployments.
- Delivered high-performance search capabilities using **Elasticsearch and OpenSearch**, enabling real-time filtering and investigation workflows across distributed application data and **Azure-integrated operational services**.
- Developed analytics pipelines integrating **Redshift, BigQuery, and Azure SQL** reporting flows, improving cross-team reporting capabilities and strengthening data-driven decision making.

- Strengthened security architecture using **OAuth2, OpenID Connect, JWT, AWS Cognito, and Azure AD / Microsoft Entra ID**, ensuring consistent enforcement of **RBAC** and **ABAC** policies across platform services and internal tools.
- Standardized distributed tracing with **OpenTelemetry** and built monitoring dashboards in **Datadog, Grafana, CloudWatch, and Azure Application Insights**, improving incident detection speed by over **50%**.
- Automated deployments using **Docker, Kubernetes (EKS/AKS), ECS, Helm, and Terraform**, enabling reliable CI/CD pipelines and repeatable infrastructure provisioning across multi-environment cloud delivery.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Developed **Vue.js (2.x) operational dashboards** that reduced manual support workflows and improved internal efficiency across operations teams.
- Architected backend services using **ASP.NET Core (.NET Core 2–5)** with clear domain separation, reducing cross-team dependencies and improving scalability.
- Designed RESTful services with stable payload contracts, reducing frontend regressions and improving integration reliability across releases.
- Implemented **RabbitMQ-based background processing systems**, resolving retry failures and ensuring reliable task execution under load.
- Enhanced streaming integrations using **Kafka**, solving consumer lag and poisoned message issues through partition tuning and validation strategies.
- Improved database reliability across **PostgreSQL, MySQL, and SQL Server** by enforcing migration discipline and optimizing slow queries.
- Introduced **Redis caching standards**, improving response times and reducing database contention under heavy workloads.
- Built vendor integration layers that isolated third-party inconsistencies, allowing internal systems to evolve independently.
- Implemented secure authentication using **OAuth2 and JWT**, ensuring consistent authorization enforcement across services and UI layers.
- Enhanced observability through **OpenTelemetry and Grafana dashboards**, significantly improving debugging of intermittent production issues.
- Expanded automated testing coverage using **Jest, Mocha, and Chai**, reducing regression rates during rapid deployment cycles.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed internal tools using **Vue.js (early 2.x)** to visualize workflows and operational data, improving team productivity and transparency.
- Built backend APIs with **C#/ .NET Framework transitioning to early .NET Core**, unifying multiple upstream systems into consistent service layers.
- Implemented strict validation and error-handling strategies, preventing unstable UI flows during partial service failures.
- Designed transactional data systems across **PostgreSQL and MySQL**, eliminating partial writes and improving financial reconciliation accuracy.
- Automated reporting workflows with scheduled jobs and retry mechanisms, replacing manual processes with reliable pipelines.
- Improved performance by analyzing slow queries and introducing targeted indexing strategies, reducing reporting delays significantly.
- Built ingestion pipelines with schema validation and quarantine mechanisms, preventing corrupted data from entering production systems.
- Introduced **MongoDB for flexible document storage**, improving handling of semi-structured metadata alongside relational systems.
- Implemented caching with **Redis**, improving read-heavy API performance and stabilizing backend load.

- Delivered search capabilities using **Elasticsearch**, enabling fast querying without overloading transactional databases.

Junior Software Developer

October 2014 to February 2016

Centric Tech Inc. - New Jersey, United States

- Developed internal admin tools using **server-rendered UI and early JavaScript-enhanced SPA patterns**, improving usability of internal workflows.
- Built **REST-style APIs** using **C#/.NET Framework**, standardizing validation and error responses to reduce support overhead.
- Implemented data access layers using **SQL Server**, improving transaction reliability and reducing inconsistencies in business operations.
- Enhanced background job processing with **RabbitMQ**, introducing retry and backoff strategies to improve job stability.
- Added **Redis-based caching** for high-read workflows, reducing redundant database queries and improving application responsiveness.
- Implemented internal search and filtering with **Elasticsearch**, enabling faster lookup for operational users without increasing database load.
- Standardized build and deployment pipelines using **Jenkins**, eliminating environment-specific deployment issues.
- Added targeted unit tests for business rules using **NUnit / MSTest**, reducing regression defects in production.
- Collaborated with senior engineers to diagnose performance bottlenecks using logs and query profiling techniques, improving system efficiency.

EDUCATION

Bachelor's degree in Computer Science

2010 to 2014

Stanford University Department of Computer Science

- Built strong foundations in **algorithms, databases, and software engineering**, applying coursework through practical team projects and production-style system design.

CERTIFICATIONS

- **AWS Certified Developer – Associate (Preparation & Practical Implementation)**
- **Advanced .NET Backend Engineering & API Design**
- **Modern Frontend Architecture with Vue.js & TypeScript**